CSIT314 Software Development Methodologies

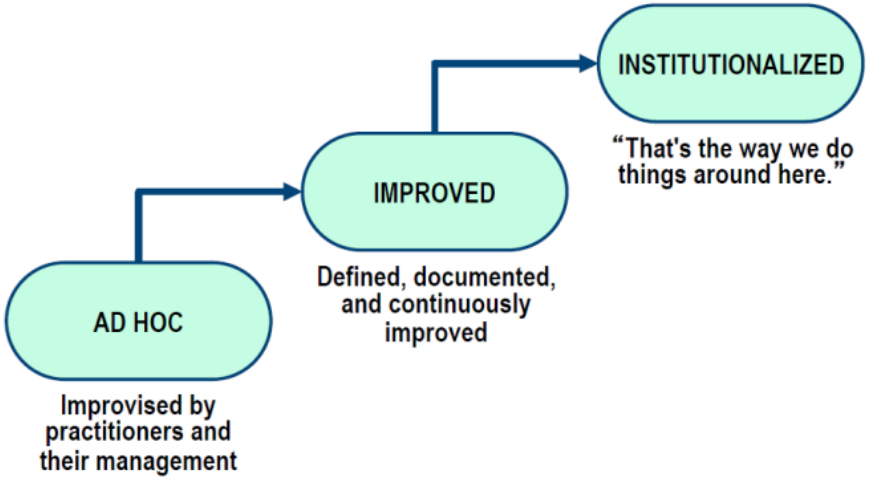
CMMI



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Definition of Process

- A set of interrelated activities, which transform inputs into outputs, to achieve a given purpose.
- The quality of a system is highly influenced by the quality of the (also inline with software design methodology) process used to acquire, develop, and maintain it.



Ad Hoc Processes

- Process descriptions are not carefully followed or enforced
  - Can be very different from people to people
- Understanding of the current status of a project is limited
- Performance is highly dependent on current participants
- XP as an example

Improved Processes

- Process descriptions are consistent with the way work actually is done
- Processes are supported visibly by management and others
- They are well controlled—process fidelity is evaluated and enforced
- There is constructive use of product and process measurement.

Institutionalised Processes

- The organization builds an infrastructure that contains effective, usable, and consistently applied processes
- The organizational culture conveys the process
- Management nurtures the culture
- Culture is conveyed through role models and recognition

Benefits of improving processes

- Better understand of how things occur in an organization
- More predictability
- People working more effective

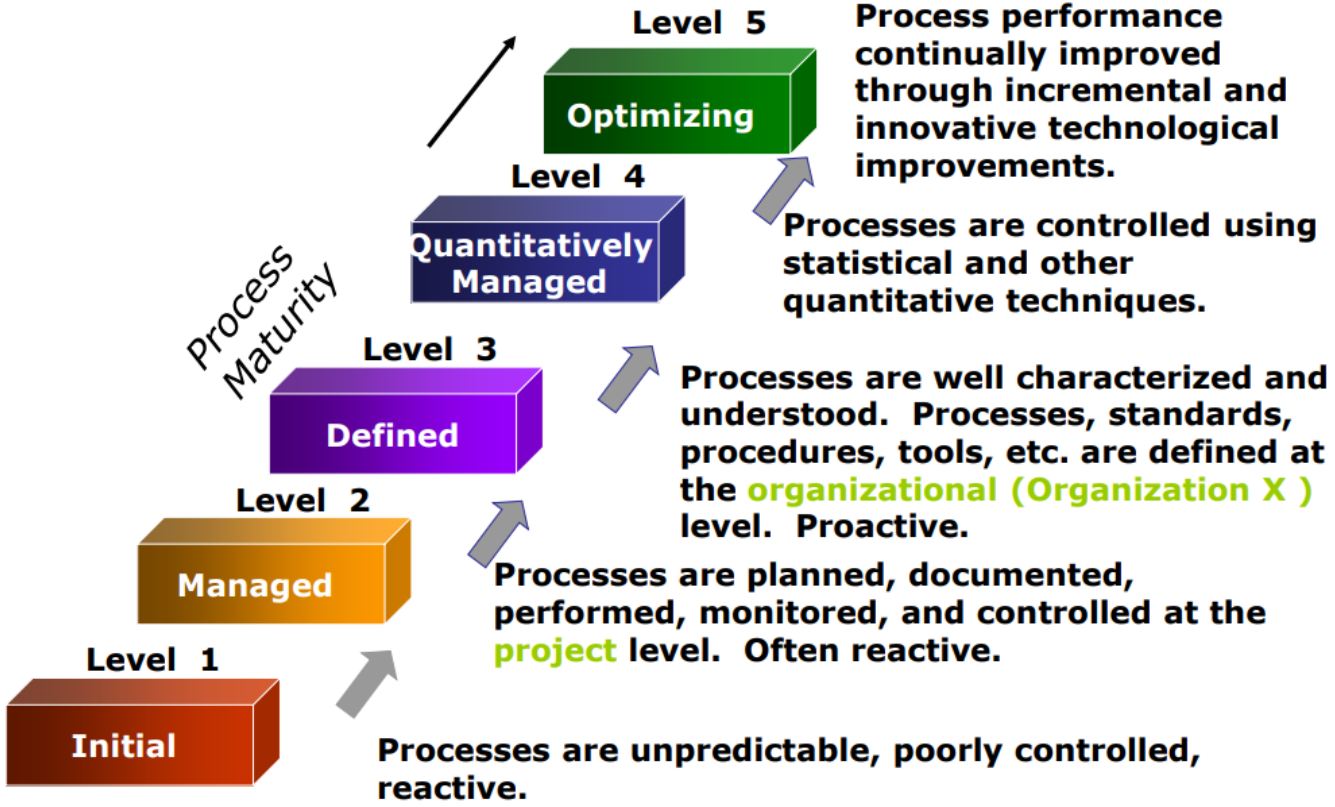
CMMI

- Capability Maturity Model Integration is a proven industry framework to improve product quality and development efficiency for both hardware and software
  - Sponsored by US Department of Defense in cooperation with Carnegie Mellon University and the Software Engineering Institute (SEI)
  - Many companies have been involved in CMMI definition such as Ericsson
  - CMMI has been established as a model to improve business results
- CMMI uses 5 levels to describe the maturity of the organization
  - Emphasis on business needs, integration and institutionalization

CMMI for Process Improvement

- Use CMMI in process improvement activities as
  - a collection of best practices
  - framework for organising and prioritizing activities
  - support for the coordination of multi-disciplined activities that might be required to successfully build a product
  - emphasize the alignment of the process improvement objectives with organizational business objectives

5 Maturity Levels



Maturity Level 1

- Maturity Level 1 deals with performed processes
  - unpredictable, poorly controlled, reactive
- The process performance may not be stable and may not meet specific objectives such as quality, cost, and schedule, but useful work can be done.

Maturity Level 2

- Maturity Level 2 deals with the Project Level and managed processes
- A managed process is a performed process
  - Planned and executed in accordance with policy
  - Employs skilled people
  - Adequate resources are available
  - Controlled outputs are produced
  - Stakeholders are involved
  - The process is reviewed and evaluated for adherence to requirements
- Processes are planned, documented, performed, monitored, and controlled at the project level
- The managed process comes closer to achieving the specific objectives such as quality, cost, and schedule.

Maturity Level 3

- Maturity Level 3 deals with defined processes (at the Organization Level)
- A defined process is a managed process that
  - Well defined, understood, deployed and executed across the entire organization
  - Project or local tailoring is allowed, however it must be based on the organization's set of standard processes and defined per the organization's tailoring guidelines

CMMI Process Areas

- Basic functions that need to be performed (within each of the 5 Maturity Levels)
  - With a set of goals and practices
- CMMI process areas are classified into four groups:
  - Project Management
  - Process Management
  - Engineering
  - Support

https://www.tutorialspoint.com/cmmi/cmmi-process-areas.htm

Project Management

- Quantitative Project Management
  - *quantitatively* manage the project's defined process to achieve the project's established quality and process-performance objectives
- Integrated Project Management
  - establish and manage the project and the involvement of the relevant stakeholders according to an integrated and defined process that is tailored from the organization's set of standard processes.

Project Management

- Risk Management
  - to identify potential problems before they occur so that risk-handling activities can be planned and invoked as needed across the life of the product or project to mitigate adverse impacts on achieving objectives
- Project Planning
  - establish and maintain plans that define project activities.

Project Management

- Project Monitoring & Control
  - provide an understanding of the project's progress so that appropriate corrections can be taken when the project's performance deviates significantly from the plan.
- Supplier Agreement Management
  - manage the acquisition of products from suppliers for which there exists a formal agreement.

Process management

- Organisational Innovation & Deployment
  - incremental and innovative improvements support the organization's quality and process-performance objectives
- Organisational Process Performance
  - a quantitative understanding of the performance, and to provide the process performance data, baselines, and models to quantitatively manage the organization's projects.

Process management

- Organisational Process Focus
  - plan and implement organizational process based on the current strengths and weaknesses
- Organisational Process Definition
  - establish and maintain a usable set of organizational process assets
- Organizational Training
  - develop the skills and knowledge of people so they can perform their roles effectively and efficiently

Engineering

- Requirements Development
  - to produce and analyze customer, product, and product-component requirements
- Technical Solution
  - design, develop, and implement solutions to requirements. Solutions, designs, and implementations either singly or in combination as appropriate
- Product Integration
  - assemble the product from the product components, ensure that the product, as integrated, functions properly, and deliver the product.

Engineering

- Verification & Validation
  - ensure that selected work products meet specified requirements
  - fulfills its intended use when placed in its intended environment
- Requirements Management
  - manage the requirements of the project's products and product components and to identify inconsistencies between those requirements and the project's plans and work products

Support

- Causal Analysis & Resolution
  - identify causes of defects and take action to prevent them from occurring in the future
- Decision Analysis & Resolution
  - analyze possible decisions and evaluate alternatives against established criteria

Support

- Measurement & Analysis
  - to support management information needs
- Process & Product Quality Assurance
  - provide objective insight into processes and associated work products
- Configuration Management
  - configuration identification, control, status accounting, and audits

CMMI Process Areas

Maturity Level	Project Managment	Engineering	Process Management	Support
5 Optimizing			Organizational Innovation & Deployment	Causal Analysis & Resolution
4 Quantitatively Managed	Quantitative Project Mngt		Organizational Process Performance	
3 Defined	Integrated Project Mngt Risk Management	Requirements Development Technical Solution Product Integration Verification Validation	Organizational Process Focus Organizational Process Definition Organizational Training	Decision Analysis & Resolution
2 Managed	Project Planning Project Monitoring & Control Supplier Agreement Mngt	Requirements Mngt		Measurement & Analysis Process & Product Quality Assurance Configuration Mngt
1 Initial				

CMMI & Agile

- Backlog grooming: common agile technique used by Scrum teams to produce a prioritized backlog of epics and user stories before and during a sprint
  - Project planning
  - Requirement development
  - Requirement management

CMMI & Agile

- User stories
  - Requirement development
  - Requirement management
  - Verification
  - Integrated project management

CMMI & Agile

- Team estimating/Planning : an agile estimation technique that establishes relative sizing using story points and rough order of magnitude estimation
  - Requirement management
  - Risk management
  - Project planning

CMMI & Agile

- Daily standup/Daily Scrum: Daily Standup Meeting is used as a way to identify issues and risks earlier than a traditional project (“fail fast”), and to increase collaboration between agile team members
  - Risk management
  - Project monitoring & control
  - Integrated Project management

CMMI & Agile

- Pair programming: usually increases the initial cost of programing, but it more than pays for itself in increased code quality.
  - Requirement development
  - Verification
- Continuous integration:
  - Verification
  - Product integration
  - Configuration management

CMMI Certification & Appraisal

- An organization gets accredited to certification level i according to CMMI if all process areas are governed that belong to maturity level 1 to i
- The certification process can be complicated and lengthy and is therefore rather suitable for big organizations
- Roughly estimated, around 2-3 years are necessary to go from maturity level i to i+1.

https://cmmiinstitute.com/learning/appraisals/results